

Rensselaer Polytechnic Institute
Department of Electrical, Computer, and Systems Engineering
ECSE-4040 - Digital Electronics
Spring Term 2026

Lab. 3-Inverter Topologies, V_{OL} and V_{OH}

Issued: 2/4/2026

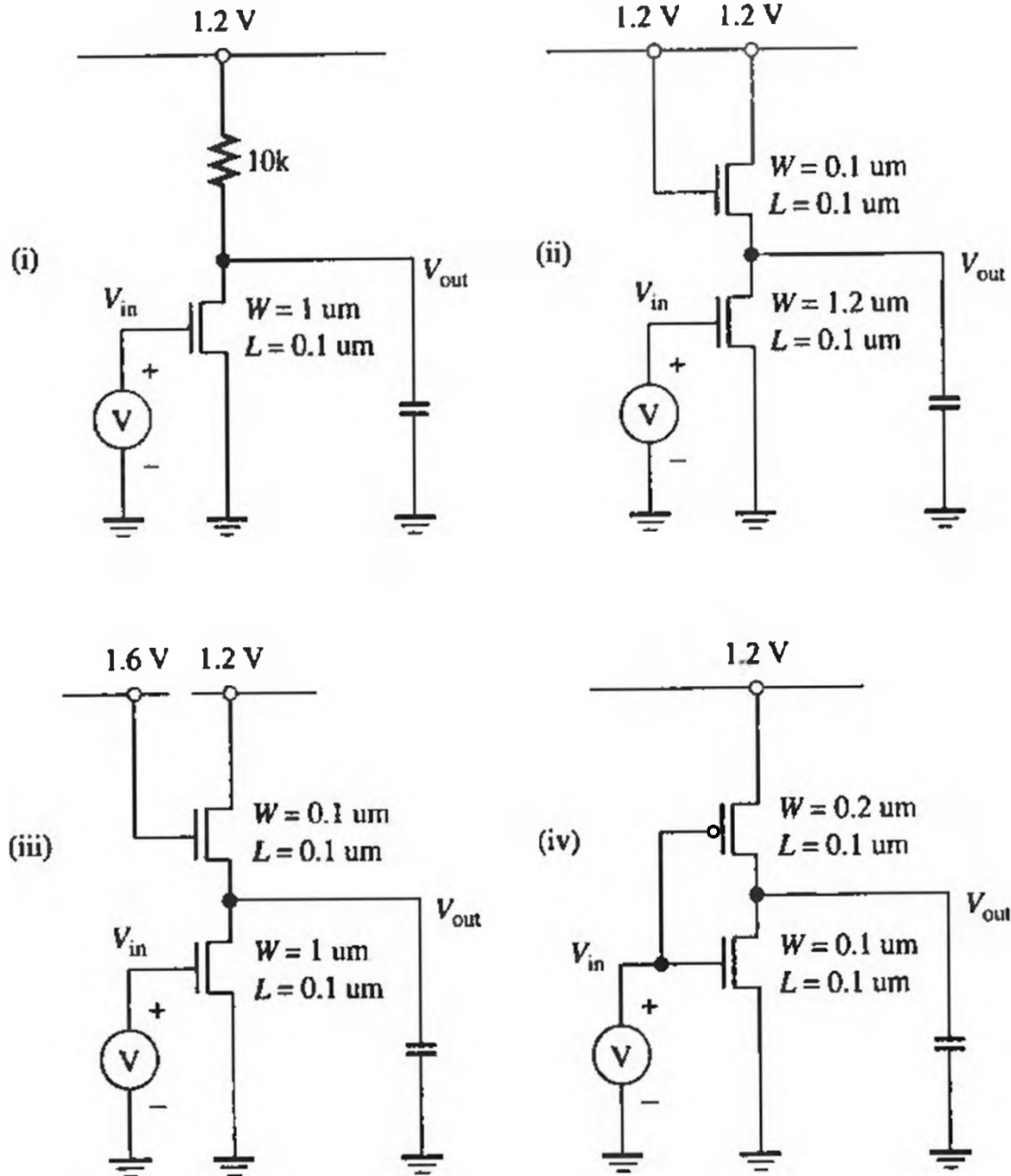
Due: 2/6/2026

Given

Below is a table of process parameters for a generic 0.13 μm technology.

0.13 μm Process technology parameters

Typical Technology Parameters	Symbol	NMOS	PMOS
Gate thickness	t_{ox} [Å]	22	22
Poly doping level	N_D [cm^{-3}]	3×10^{20}	—
Poly doping level	N_A [cm^{-3}]	—	3×10^{20}
Substrate doping level	N_A [cm^{-3}]	3×10^{17}	—
N-well doping level	N_D [cm^{-3}]	—	3×10^{17}
Number of surface state charges	N_{ss} [cm^{-2}]	6×10^{11}	6×10^{11}
Electron Mobility	μ_n [cm^2/Vs]	270	—
Hole Mobility	μ_p [cm^2/Vs]	—	70
Critical Field x Length	$E_C L$ [V]	0.6	2.4
Saturation velocity	v_{sat} [cm/s]	8×10^6	8×10^6
Gamma	γ [$\text{V}^{1/2}$]	0.2	0.2
Oxide capacitance	C_{ox} [F/ cm^2]	1.6×10^{-6}	1.6×10^{-6}
Threshold voltage	V_T [V]	0.4	− 0.4
2 x Fermi voltage	$2 \phi_F $ [V]	0.88	0.88
Channel length modulation	λ	0.2	0.2



All problems assume $0.13 \mu\text{m}$ technology parameters as shown in the given table. The circuits of the figure above show different implementations of an inverter with output connected to a capacitor. For hand calculations you are expected to use short channel MOSFET equations.

Question 1: Based on Problem 4.1 (D.A. Hodges, H.G. Jackson, and R.A. Saleh, *Analysis and Design of Digital Integrated Circuits*, 3rd Ed., McGraw-Hill, 2004.)

Objectives:

- Familiarize yourself with the advantages and disadvantages of various forms of inverters.

Answer the following questions referencing the four given inverter circuits:

- a) Which one of the circuits consumes static power when the input is high?
- b) Which one of the circuits consumes static power when the input is low?
- c) V_{OH} of which circuit(s) is 1.2 V?
- d) V_{OL} of which circuit(s) is 0 V?
- e) The proper functionality of which circuit(s) depends on the size of the devices?

Question 2: Based on Problem 4.2(D.A. Hodges, H.G. Jackson, and R.A. Saleh, *Analysis and Design of Digital Integrated Circuits*, 3rd Ed., McGraw-Hill, 2004.)

Objectives:

- 1. Perform hand calculations to find the output voltage of various inverters.
- 2. Use LTspice to verify the behavior of the circuit by comparing hand calculations to the DC operating voltage.

Calculate the V_{OH} and V_{OL} of the four inverters in the given figure by the means below:

- a. By hand calculations using the 0.13 μm process parameters.
 - i. Use short-channel MOSFET equations where applicable.
- b. By LTspice by performing a transient simulation on all four circuits.
 - i. Use the updated **SPICE level 49 NMOS & PMOS.txt** file provided on LMS in the Labs folder.
 - ii. Transient simulation duration is to be set to 10 μs .
 - iii. Place a 1 pF load capacitor out the output node of each inverter.
 - iv. Read and follow the LTspice instructions in the following section to ensure correct simulation.

LTSpice:

To simulate correctly, we need to make the following accommodations in LTSpice:

- To achieve correct V_{OH} and V_{OL} results with LTSpice, due to MOSFET model issues it is necessary to double all W and L values from the figure.
 - Ex: length of 100nm is doubled to 200nm, 1 μm width is doubled to 2 μm .
 - The same W/L ratios are to be preserved in simulation despite the sizing change.
- If the output initial voltage is undefined, the initial voltage may be set to a voltage outside of V_{OL} and V_{OH} , making it impossible to simulate V_{OL} or V_{OH} .
 - Ex: If a circuit can only pull up to 1V, but the simulator initially sets the output to 1.1V, V_{OH} will improperly be reported at 1.1V.
 - The following SPICE Directive syntax is used to set an initial condition:
 - **.ic V(Vout) = 0**
 - In the above, Vout is initially set to 0 V.
 - Initial output voltage when testing VOL must be set to 1.2V
 - Initial output voltage when testing VOH must be set to 0V
 - If simulating all 4 inverter circuits at once, all 4 outputs must individually be set to an initial value.
- When defining an initial voltage, we can no longer use a DC operating point simulation.
 - The output reported will be the same as your initial value.
 - We must use a transient simulation.

We can make our lives simpler by using parameters (variables).

- This is useful to set the same Vin for all 4 inverters at once.
- This is done with a .param command and curly brackets
 - **.param vin = 0**
 - The above creates the **vin** parameter and sets it to 0V.

Below are examples of key points for simulation in this lab:

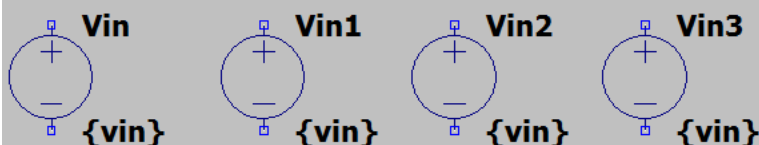
- Don't forget the NMOS and PMOS models
- Set a 100 μs long transient simulation
- Set all output nodes to an initial voltage, in this example the **vinit** parameter
- Set parameters
- Have components read parameters by putting the parameter name in {} brackets.

```
.lib C:\rpi teaching\digital electronics\130nm_bulk.txt
```

```
.tran 100u
```

```
.ic V(Vout1 )=vinit V(Vout2 )=vinit V(Vout3)=vinit V(Vout4)=vinit
```

```
.param vin = 0 vinit = 0
```



Report Submission

Submit your work on LMS under **Labs**

- Show your work.
 - Include images of your hand calculations with work and final results.
 - Include screenshots of LTspice with all circuit schematics with all output plots, clearly showing all four V_{OH} and V_{OL} values.
- Due date: Friday at 11:59pm.